

UCS301: Data Structures

L	T	P	Cr
3	0	2	4.0

Course Objective: To become familiar with different types of data structures and their applications.

Syllabus

Analysing algorithms: Basics of algorithm and its analysis, Complexity classes, order arithmetic, Time and space trade-off in algorithms. Practical implications of inefficient algorithms on energy consumption, need of minimizing computational overhead for sustainability.

Linear Data Structures: Arrays, Strings and string processing, Linked lists (Singly, Doubly, Circular), Abstract data types, their implementation and applications: Stacks (using Arrays and Linked-list), Queues (using Arrays and Linked-list), Hash tables: Hash functions, collision resolution techniques, Strategies for choosing the appropriate data structure.

Searching and Sorting: Linear Search, Binary Search. Introduction to internal and external sort, Bubble Sort, Selection Sort, Insertion Sort, Shell Sort, Quick Sort, Merge Sort, Counting Sort, Radix Sort.

Trees and their applications: Introduction to binary tree, tree traversal algorithms, Binary search tree, AVL Tree, B Tree etc. and common operations on these trees. Spatial data structures (e.g., Quadrees, R-trees etc.) for sustainable urban planning and waste management. Heap, Heap Sort, Priority Queue using Heap.

Graphs and their applications: Graph Terminology and its representation, Depth and breadth first traversals, Shortest-path algorithms (Dijkstra and Floyd), Data Structures for Disjoint Sets, Minimum spanning tree (Prim and Kruskal). Applications of graphs in smart grid management, waste collection routes, recycling networks etc.

Laboratory Work

Implementation of various data structures such as Arrays, Stacks, Queues, Lists, Binary tree traversals, BST, AVL trees, Graphs traversals, Sorting and Searching techniques.

Course Learning Objectives (CLO)

The students will be able to:

1. Understand the fundamental data structures, their implementation and some of their standard applications.
2. Select and implement appropriate searching and sorting techniques for solving a problem based on their characteristics.
3. Apply tree and graph data structures for specific applications.
4. Design and analyse algorithms using appropriate data structures for real-world problems.

Text Books

1. Introduction to Algorithms, Cormen H. T., Leiserson E. C., Rivest L. R., and Stein C, MIT Press, 3rd ed., 2009
2. Data Structures, Algorithms and Applications in C++, Sahni S., Universities Press 2nd ed. 2005.

Reference Books

1. Data Structures and Algorithms Made Easy, Karumanchi N., Career Monk Publications, 5th ed., 2017
2. Data structures and algorithms in C++, Adam Drozdek, 4th edition.